

LABORATORY RECORD

Course	Full Stack Web Development	Date	22-09-2026
Course Code	23CM4121	Exp. No.	2
Student Name	Badana Pranathi	Roll No.	A24126552071

1. TITLE

Design and Implementation of a College Information Portal Using HTML5 Text Formatting, Lists, Tables, Multimedia, Semantic Elements and Forms

2. AIM

To develop a static College Information Portal web page demonstrating HTML5 text formatting tags, superscripts, subscripts, ordered and unordered lists, structured tables, images, hyperlinks, a student registration form with multiple input types, HTML5 semantic elements, and embedded audio/video.

3. OBJECTIVE

The objectives of this experiment are:

- To practice document structuring and text formatting with HTML5 (bold, italic, underline, sub, sup).
- To construct ordered and unordered lists for courses and facilities.
- To create a structured table displaying student academic details along with images.
- To design a registration form using text, email, password, radio, checkbox, dropdown, and textarea inputs.
- To apply HTML5 semantic elements such as header, nav, section, article and footer.
- To embed multimedia content using the audio and video elements.
- To explore new HTML5 input types such as date, number, color and range.

4. THEORY

HTML5 is the backbone of modern web development. It provides semantic elements for meaningful page structure, inline text-formatting tags for emphasis and scientific notation, list elements for organising content, tables for tabular datasets, the `img` and `a` elements for media and navigation, the `form` element with a rich set of input types for capturing user data, and native audio/video elements for embedding multimedia without external plugins.

Application Flow: HTML5 Structure -> Text Formatting & Lists -> Student Table & Images -> Registration Form -> Semantic Layout & Multimedia

5. ALGORITHM / PROCEDURE

1. Create an HTML5 document with the title 'College Information Portal'.
2. Part A: Display the college heading and an introductory paragraph using bold and italic tags; format text with sub and sup for H₂O and x₂; build an ordered list of courses and an unordered list of facilities; add hyperlinks with `target='_blank'`.

3. Part B: Construct a bordered, padded table with headers Roll No, Name, Branch and Email, and populate it with student records; embed the college logo and campus images with defined width and height.
4. Part C: Implement a student registration form containing text, email and password fields, radio buttons for gender, checkboxes for hobbies, a dropdown for branch, a textarea for comments, and submit/reset buttons.
5. Part D: Structure the page using semantic elements header, nav, section, article and footer; embed an audio element for the college anthem and a video element for the campus tour; add HTML5 input types for date, number, color and range.
6. Save the file as index.html and open it in a web browser to verify rendering.
7. Test each section (formatting, lists, table, form, semantic layout, multimedia) for correct display and behaviour.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html>

<head>
    <title>College Information Portal</title>
</head>

<body>
<b><h1>PART A</h1></b>
    <h1>College Information Portal</h1>
    <h2>About Our College</h2>
    <p>
        Welcome to <b>Anil Neerukonda Institute of Technology and Scinces</b>. Our college provides
        <i>quality education</i> and helps students build a successful career.
    </p>
    <h2>MFML COURSE TOPICS</h2>
    <p>
        <b>ML with AI</b><br>
        <i>Linear regression</i><br>
        <u>MATRICES</u><br>
 $H^2O$       <br>
 $x^2$ 
    </p>
    <h2>Courses Offered</h2>
    <ol>
        <li>Computer Science</li>
        <li>Artificial Intelligence</li>
        <li>Electronics</li>
    </ol>
    <h2>Facilities</h2>
    <ul>
        <li>Library</li>
        <li>Computer Lab</li>
        <li>Sports</li>
    </ul>
    <h2>Useful Links</h2>
    <a href="https://www.anits.edu.in" target="_blank">College Website</a>
    <br><br>
    <a href="https://www.google.com" target="_blank">Google</a>
    <hr>
    <br>
<b><h1>PART B</h1></b>
    <h1>College Information Portal</h1>
    <h2>Student Details</h2>
    <table border="1" cellpadding="10" cellspacing="0">
        <tr><th>Roll No</th><th>Name</th><th>Branch</th><th>Email</th></tr>
        <tr><td>A24126552071</td><td>Badana Pranathi</td><td>CSE (AI & ML)</td><td>badanapranathi.24.csm@anits.edu.in</td></tr>
        <tr><td>A24126552074</td><td>Boddapu Sailakshmi</td><td>CSE (AI & ML)</td><td>boddapusailakshmi.24.csm@anits.edu.in</td></tr>
        <tr><td>A24126552087</td><td>Karri Rama</td><td>CSE (AI & ML)</td><td>karrirama.24.csm@anits.edu.in</td></tr>
        <tr><td>A24126552099</td><td>Samhavaya Subanvitha</td><td>CSE (AI & ML)</td><td>samhavyasubanvitha24.csm@anits.edu.in</td></tr>
        <tr><td>A24126552125</td><td>Vanisree Meesala</td><td>CSE (AI & ML)</td><td>vanisreemeesala.csm@anits.edu.in</td></tr>
    </table>
    <br>
    <h2>College Logo</h2>
    
```

```

<br><br>
<h2>College Campus</h2>

<hr>
<br>
<b><h1>PART C</h1></b>
<h1>Student Registration Form</h1>
<form>
  <label>Name:</label><br>
  <input type="text" name="name"><br><br>
  <label>Email:</label><br>
  <input type="email" name="email"><br><br>
  <label>Password:</label><br>
  <input type="password" name="password"><br><br>
  <label>Gender:</label><br>
  <input type="radio" name="gender" value="Male"> Male
  <input type="radio" name="gender" value="Female"> Female
  <input type="radio" name="gender" value="Other"> Other
  <br><br>
  <label>Hobbies:</label><br>
  <input type="checkbox" name="hobby" value="Reading"> Reading
  <input type="checkbox" name="hobby" value="Sports"> Sports
  <input type="checkbox" name="hobby" value="Listening to Music">Listening to Music
  <input type="checkbox" name="hobby" value="Dancing">Dancing
  <input type="checkbox" name="hobby" value="Drawing">Drawing
  <br><br>
  <label>Branch:</label><br>
  <select name="branch">
    <option>CSE</option>
    <option>CSE (AI & ML)</option>
    <option>CSE (Data Science)</option>
    <option>ECE</option>
    <option>EEE</option>
    <option>Mechanical</option>
    <option>Civil</option>
  </select>
  <br><br>
  <label>Comments:</label><br>
  <textarea name="comments" rows="5" cols="30"></textarea>
  <br><br>
  <input type="submit" value="Submit">
  <input type="reset" value="Reset">
</form>
<hr>
<br>
<b><h1>PART D</h1></b>
<header>
  <h1>College Information Portal</h1>
</header>
<nav>
  <a href="#">Home</a> |
  <a href="#">Courses</a> |
  <a href="#">Contact</a>
</nav>
<br>
<section>
  <article>
    <h2>About Our College</h2>
    <p>
      Our college provides quality education and modern facilities
      for students to achieve academic excellence.
    </p>
  </article>
</section>

```

```
<br>
<h2>College Anthem</h2>
<audio controls>
  <source src="Chanakya-By-Rishabh-Sharma.mp3" type="audio/mpeg">
  Your browser does not support the audio element.
</audio>
<br><br>
<h2>College Campus Video</h2>
<video width="500" controls>
  <source src="video.mp4" type="video/mp4">
  Your browser does not support the video tag.
</video>
<br><br>
<h2>HTML5 Input Types</h2>
<form>
  <label>Email:</label><br>
  <input type="email"><br><br>
  <label>Date of Birth:</label><br>
  <input type="date"><br><br>
  <label>Age:</label><br>
  <input type="number"><br><br>
  <label>Favorite Color:</label><br>
  <input type="color"><br><br>
  <label>Programming Skill Level:</label><br>
  <input type="range" min="0" max="100"><br><br>
</form>
<footer>
  <p>&copy; 2026 College Information Portal</p>
</footer>
</body>
</html>
```

7. CODE EXPLANATION

Code / Syntax	Explanation
<code>, <i>, <u></code>	Applies bold, italic, and underline styling to headings and key terms.
<code><sub>, <sup></code>	Renders subscript (H ₂ O) and superscript (x ²) notation within a paragraph.
<code>, , </code>	Creates a numbered list of courses offered and a bulleted list of campus facilities.
<code><table border cellpadding cellspacing></code>	Builds a bordered, padded table displaying student roll number, name, branch and email.
<code></code>	Embeds the college logo and campus image with defined display dimensions.
<code><a href target="_blank"></code>	Hyperlinks that open the college website and Google in a new browser tab.
<code><input type="text/email/password"></code>	Collects the registrant's name, email address and a masked password field.
<code><input type="radio" name="gender"></code>	Radio group restricting gender selection to one mutually exclusive option.
<code><input type="checkbox"></code>	Checkbox group allowing multiple hobby selections simultaneously.
<code><select>, <option></code>	Dropdown list used to choose the student's branch.
<code><textarea rows cols></code>	Multi-line input box for free-form comments/feedback.
<code><header>, <nav>, <section>, <article>, <footer></code>	HTML5 semantic elements that structure the page into a header, navigation bar, main content article, and footer.
<code><audio controls><source></audio></code>	Embeds a playable audio track (college anthem) with native browser controls.
<code><video width controls><source></video></code>	Embeds a playable video (campus tour) with native browser controls.
<code><input type="date/number/color/range"></code>	HTML5 input types providing native date pickers, numeric steppers, colour pickers and slider controls.

8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Text formatting & About section	Inspect bold, italic, underline, sub/superscript	Bold college name, italic phrase, underlined term, H ₂ O and x ² render correctly	Pass
TC-02	Ordered & unordered lists	Check Courses and Facilities sections	Courses numbered 1-3; Facilities shown as bullet points	Pass

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-03	Hyperlinks	Click College Website and Google links	Both links open target URL in a new browser tab	Pass
TC-04	Student details table	Inspect table structure and data	Table displays Roll No, Name, Branch, Email for 5 students with visible borders	Pass
TC-05	Images	Inspect logo and campus image tags	Images render at specified width/height (placeholder icon shown if file missing)	Pass
TC-06	Registration form	Fill name, email, password, gender, hobbies, branch, comments	All fields accept input; radio and checkbox groups behave correctly; dropdown lists branches	Pass
TC-07	Submit / Reset buttons	Click Submit and Reset	Reset clears form fields; Submit triggers form submission	Pass
TC-08	Semantic layout	Inspect header, nav, section, article, footer	Page structured with a header, navigation links, an article inside a section, and a footer	Pass
TC-09	Audio & video playback	Play embedded audio and video	Native browser controls appear; media plays when a valid source file is present	Pass
TC-10	HTML5 input types	Inspect date, number, color, range inputs	Browser renders native date picker, numeric stepper, colour swatch and slider	Pass

9. OUTPUT

Output: College Information Portal - Rendered in Browser

Part A - Text Formatting, Lists and Links

PART A

College Information Portal

About Our College

Welcome to **Anil Neerukonda Institute of Technology and Scinces**. Our college provides *quality education* and helps students build a successful career.

MFML COURSE TOPICS

ML with AI

Linear regression

MATRICES

H₂O x²

Courses Offered

1. Computer Science
2. Artificial Intelligence
3. Electronics

Facilities

- Library
- Computer Lab
- Sports

Useful Links

[College Website](#)

[Google](#)

Part B - Student Details Table and Images

PART B

College Information Portal

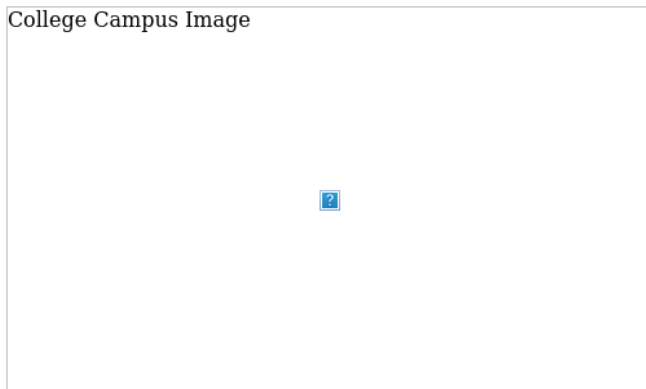
Student Details

Roll No	Name	Branch	Email
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College Logo



College Campus



Part C - Student Registration Form

PART C

Student Registration Form

Name:

Email:

Password:

Gender:

☐ Male ☐ Female ☐ Other

Hobbies:

☐ Reading ☐ Sports ☐ Listening to Music ☐ Dancing ☐ Drawing

Branch:

Comments:

Submit

Reset

Part D - Semantic Layout, Multimedia and HTML5 Inputs

PART D

College Information Portal

[Home](#) | [Courses](#) | [Contact](#)

About Our College

Our college provides quality education and modern facilities for students to achieve academic excellence.

College Anthem



College Campus Video



HTML5 Input Types

Email:

Date of Birth:

Age:

Favorite Color:



Programming Skill Level:

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10. RESULT

Thus, the College Information Portal web page demonstrating HTML5 text formatting, lists, tables, images, hyperlinks, forms, semantic elements, and multimedia was successfully designed, implemented, and verified.